

Data Flow Diagram

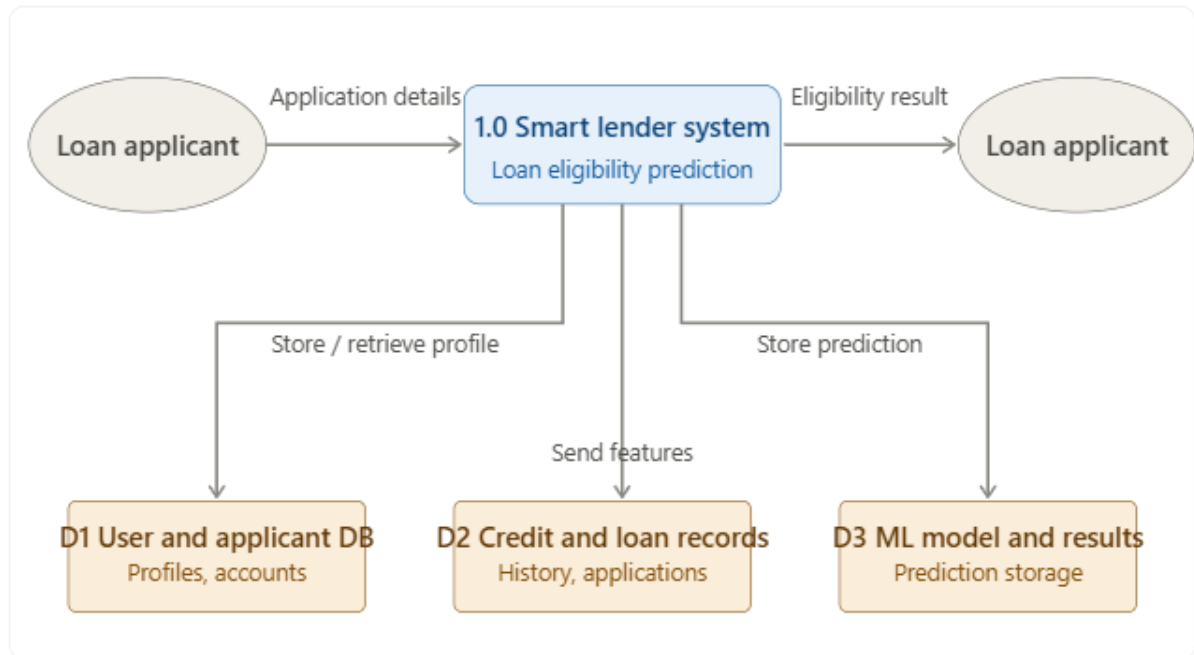
Date	11 July 2026
Team ID	SWTID-2026-1298
Project Name	Smart Lender
Maximum Marks	2 Marks

Database Architecture (Schema Specification):

The **Smart Lender** system organizes and manages information through six interconnected entities. Each entity stores specific data required for loan evaluation, machine learning prediction, and result management.

Table Name	Primary Key	Key Fields	Description & Purpose
USERS	user_id	Full name, email, password, role	Stores user account information and manages secure authentication for accessing the application.
APPLICANTS	applicant_id	user_id (FK), gender, marital_status, education, employment_status	Maintains personal and demographic details of each loan applicant.
LOAN_DETAILS	loan_id	applicant_id (FK), loan_amount, loan_term, property_area	Stores loan request information submitted by the applicant.
FINANCIAL_RECORDS	finance_id	applicant_id (FK), applicant_income, coapplicant_income, credit_history	Contains financial details used for assessing loan eligibility.
ML_MODEL	model_id	model_name, algorithm, accuracy	Stores information about the trained machine learning model used for prediction.
PREDICTION_HISTORY	prediction_id	loan_id (FK), model_id (FK), prediction_result, prediction_date	Saves prediction outcomes and maintains a history of loan approval decisions.

EXAMPLE:



Application Processing Flow:

The Smart Lender system follows a structured workflow to evaluate loan applications.

Step 1 – Secure User Access

The user signs into the application using authenticated credentials.

Step 2 – Applicant Information Collection

Personal information and financial details are entered through the application interface.

Step 3 – Data Validation

The system validates user input and prepares the information for processing.

Step 4 – Intelligent Prediction

The trained machine learning model analyzes the applicant's information and predicts the loan approval status.

Step 5 – Result Generation

The prediction outcome is displayed to the user and stored for future reference

